

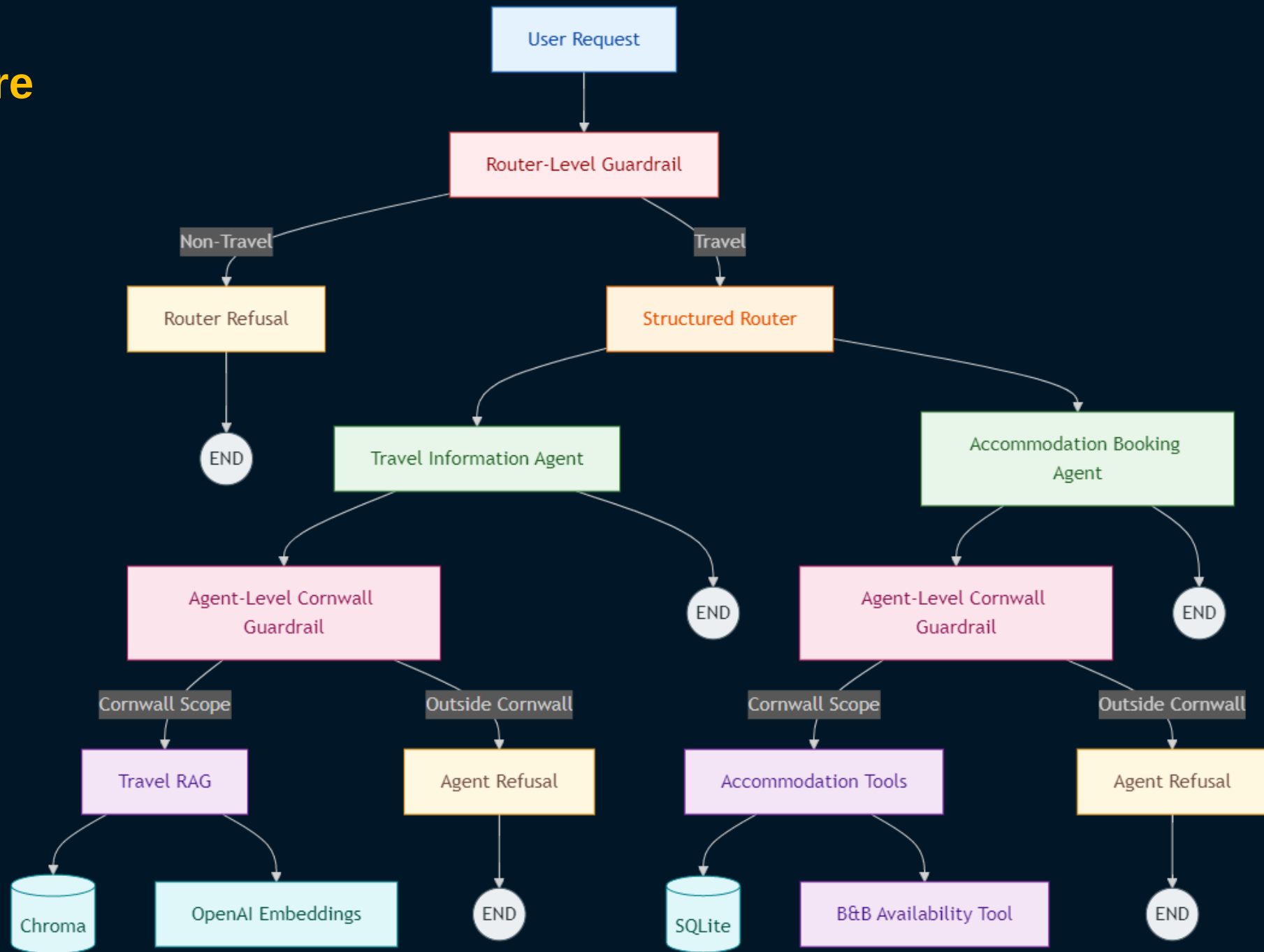
Agentic AI Engineering

***Defense-in-Depth Multi-Agent AI Platform
with
Router- and Agent-Level Guardrail***

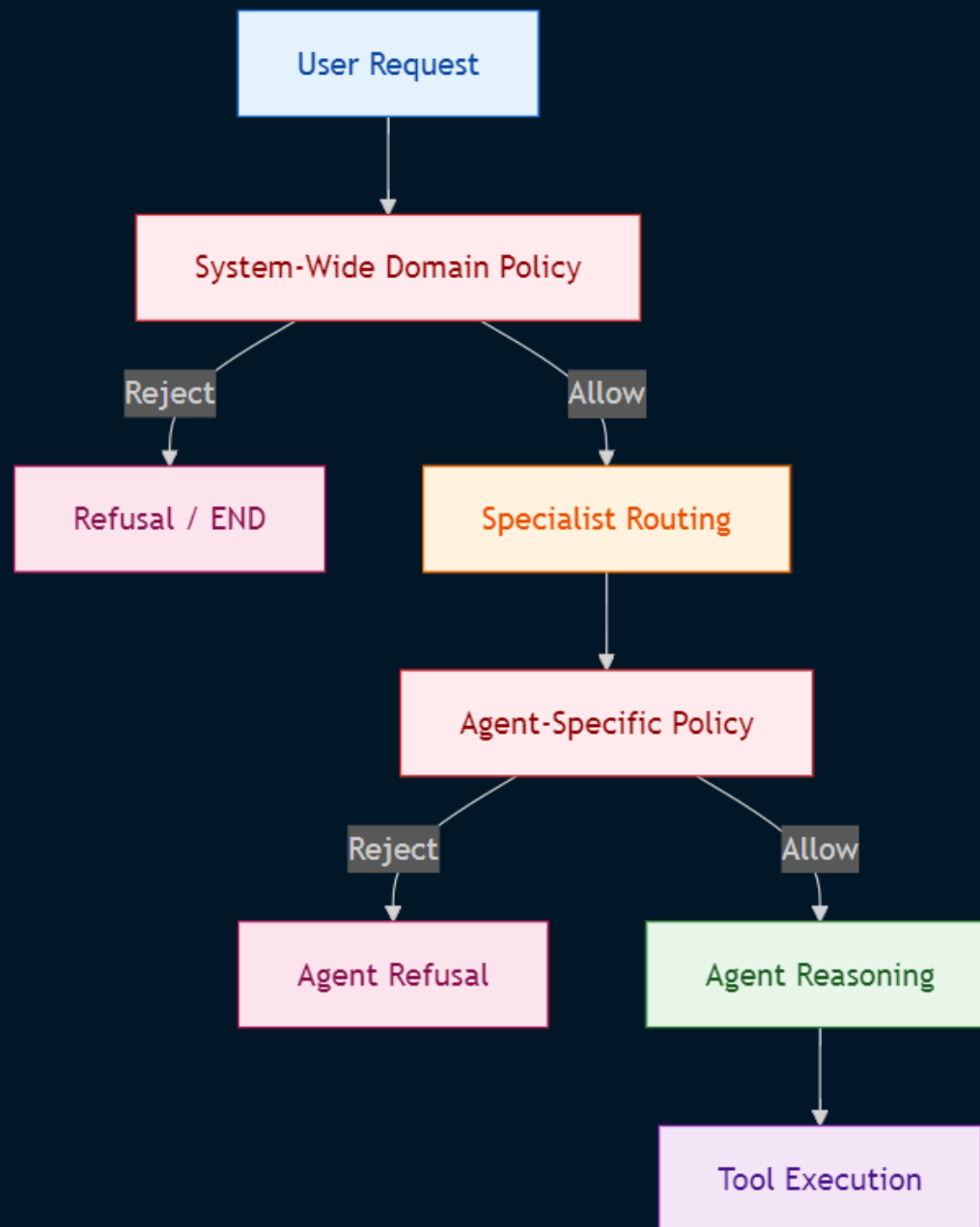
Max Montaseri

- Architected a **layered Agentic AI safety architecture** in which guardrails are enforced at both the **router level** and the **individual agent level**.
- Implemented **structured LLM policy classification with Pydantic** to enforce broad travel-domain constraints before specialist routing.
- Designed reusable **pre-model agent guardrails** that enforce stricter Cornwall-specific scope independently within travel and accommodation agents.
- Implemented a **fail-fast router guardrail** that rejects out-of-domain requests before specialist-agent and tool execution.
- Applied a **belt-and-suspenders policy architecture** so specialist agents remain protected even when reused or invoked outside the primary router workflow.
- Aligned **agent policy, RAG knowledge scope, and domain capabilities** to prevent agents from answering outside their supported geographic coverage.
- Integrated **Chroma RAG, OpenAI embeddings, SQLiteDatabaseToolkit, SQLite** and **custom tool interfaces** into specialized Agentic AI workflows.
- Combined **LangGraph checkpointing, thread management, OpenAI Responses API, and structured routing** to support stateful multi-turn execution.

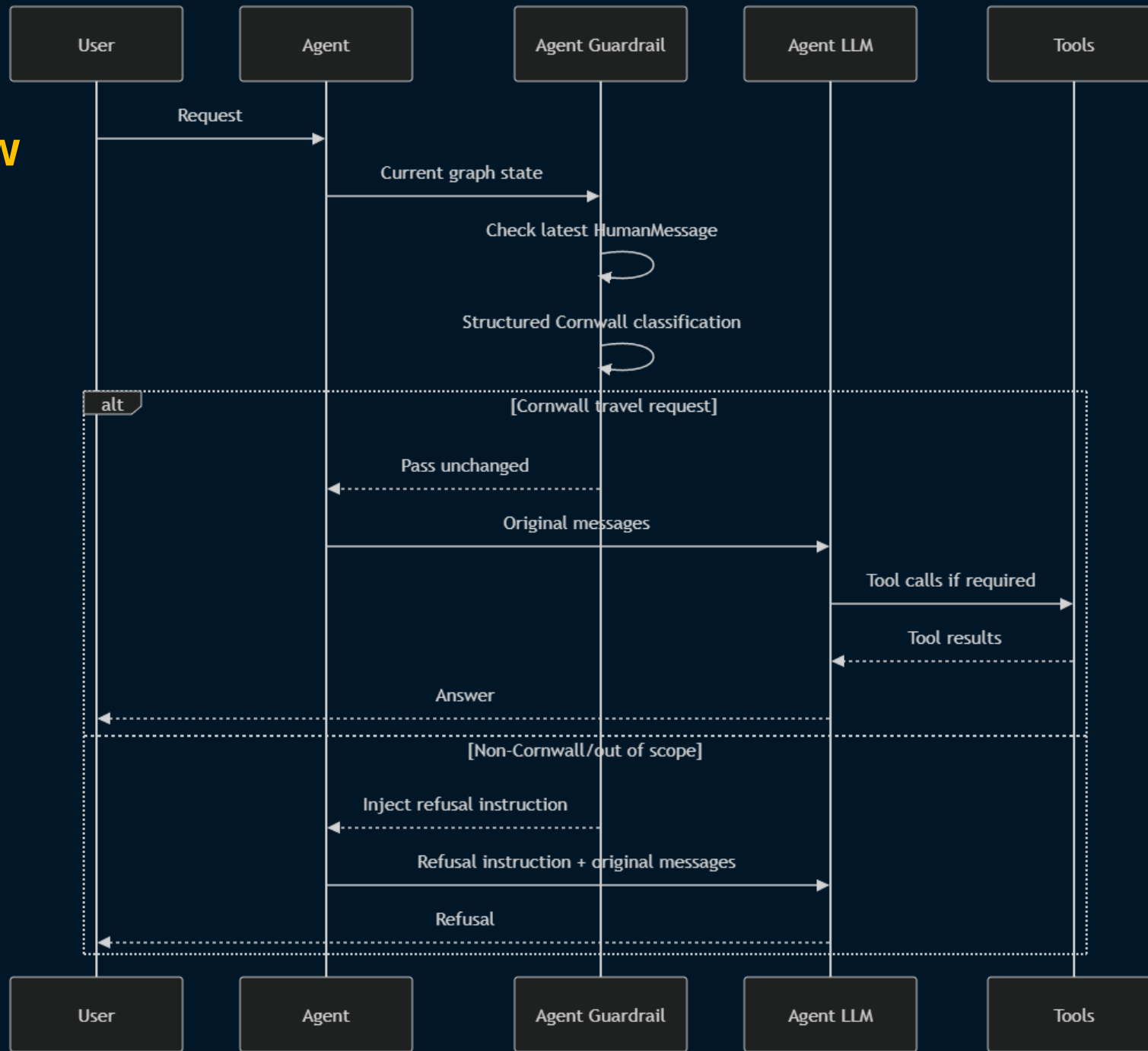
Architecture



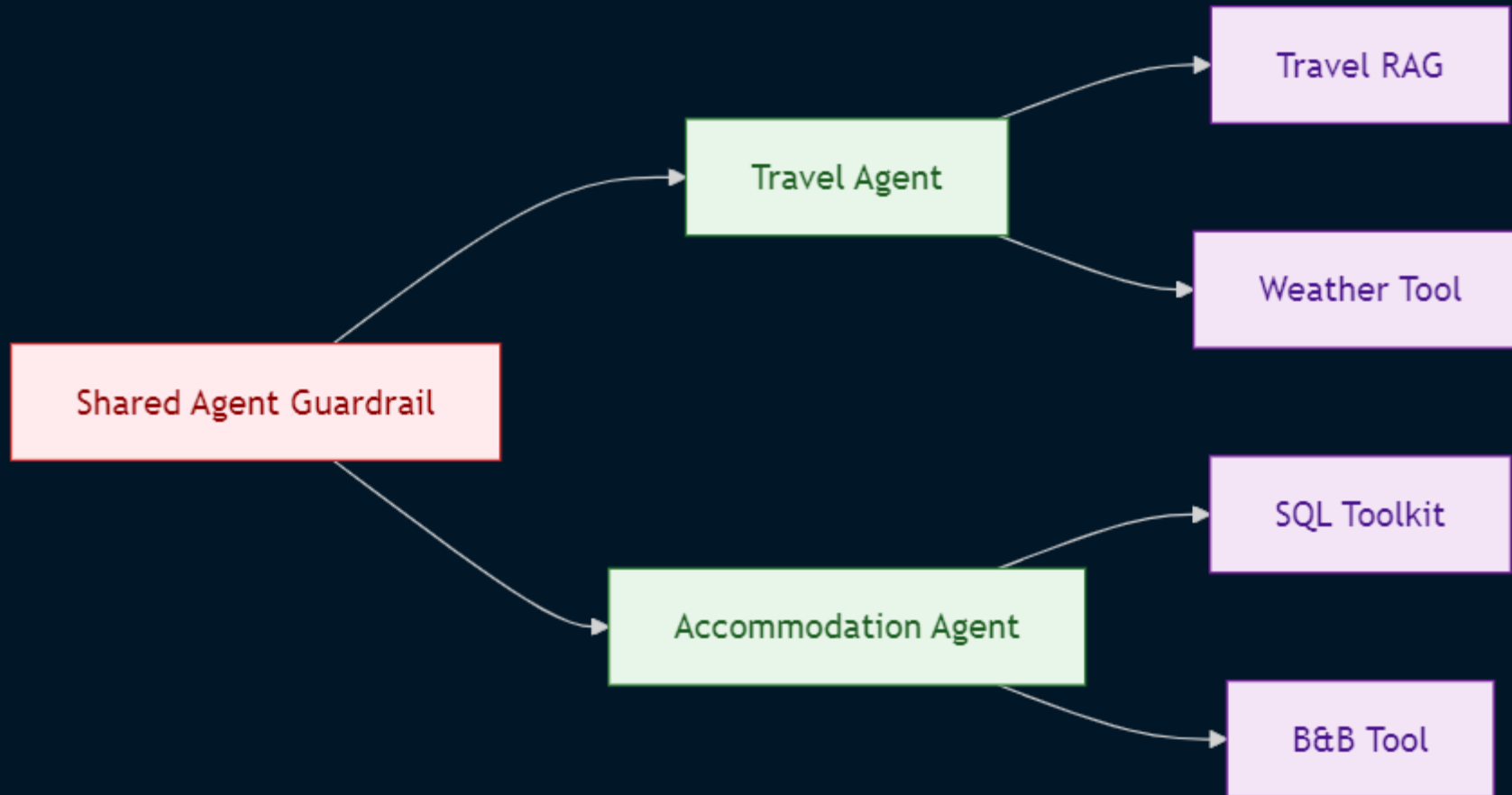
Policy Hierarchy



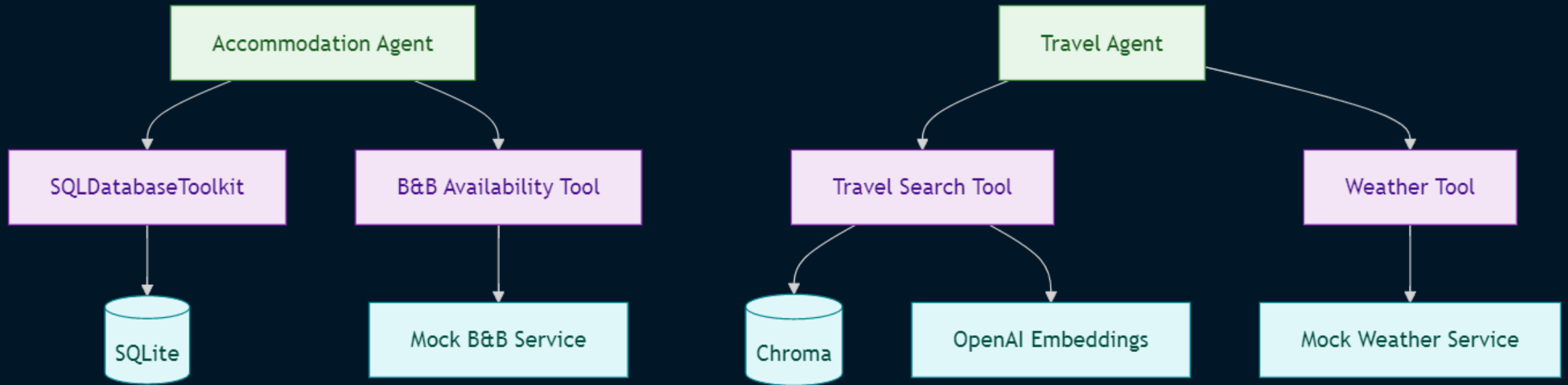
Agent-Level Guardrail Flow



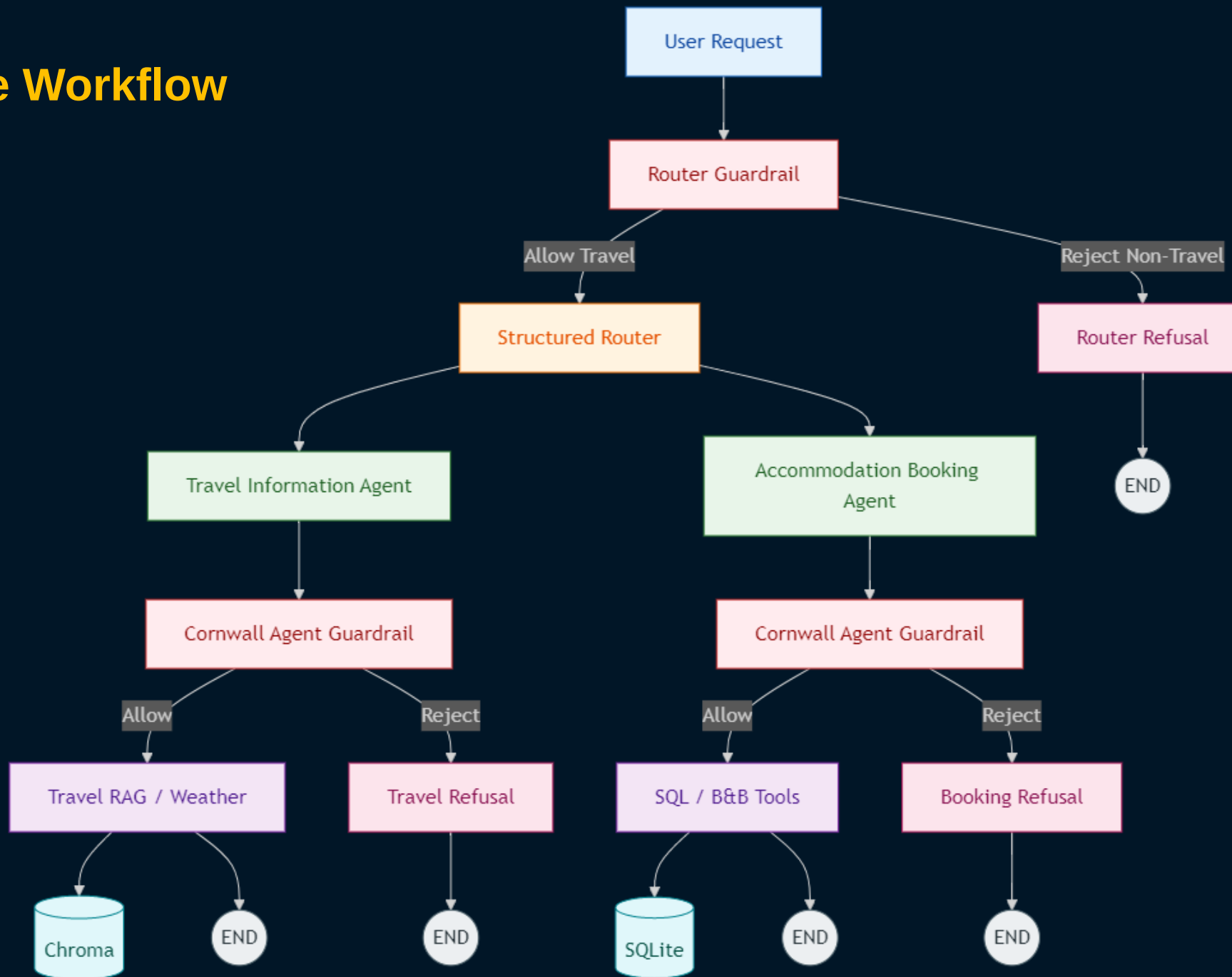
Reusable Agent Guardrail



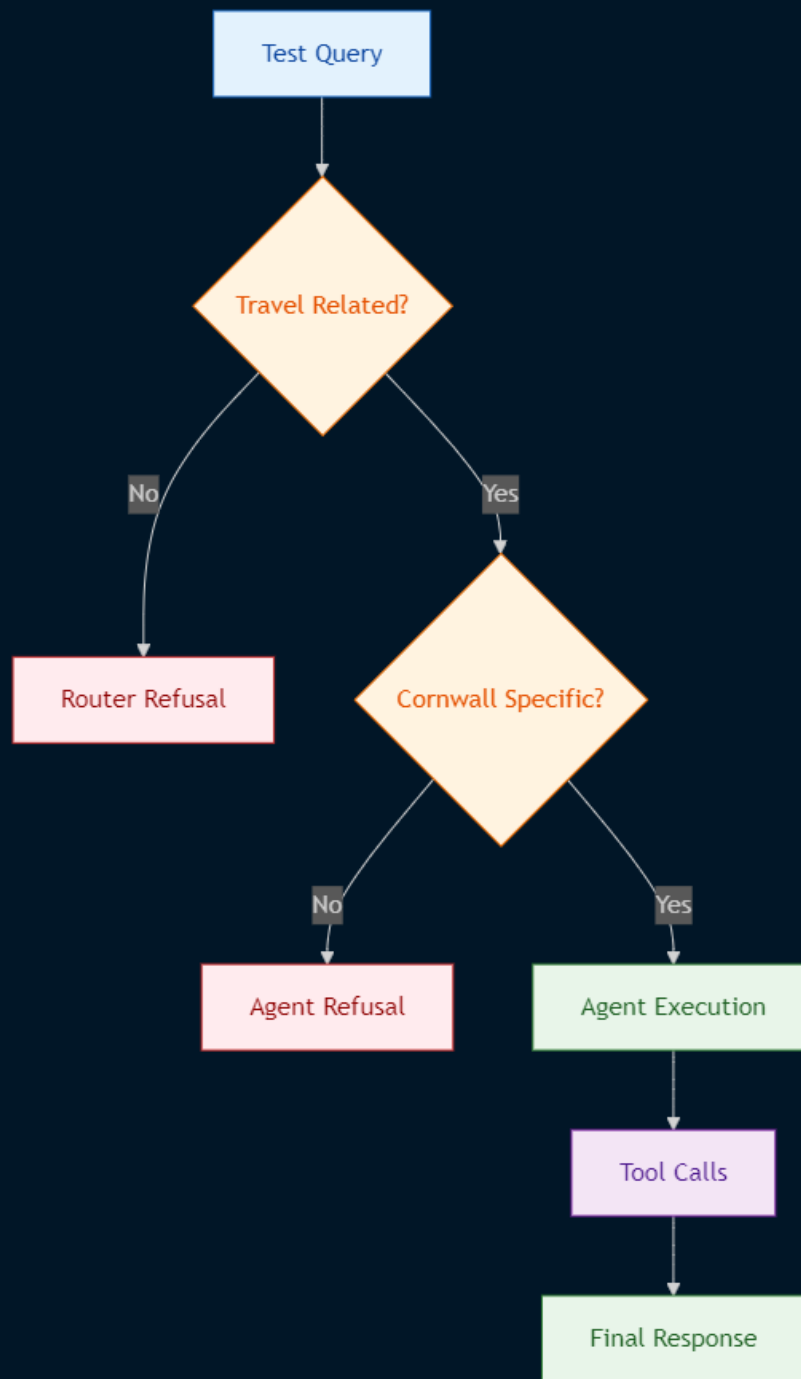
Tool Boundary



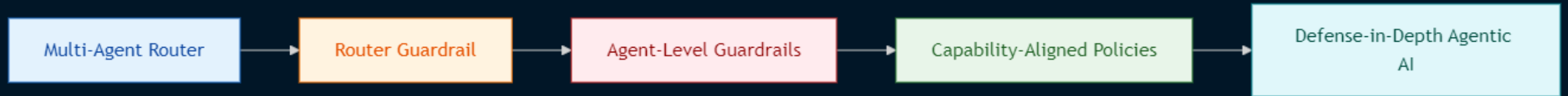
Complete Workflow



Two-Layer Test Strategy



Architectural Evolution



Production Guardrail Roadmap

